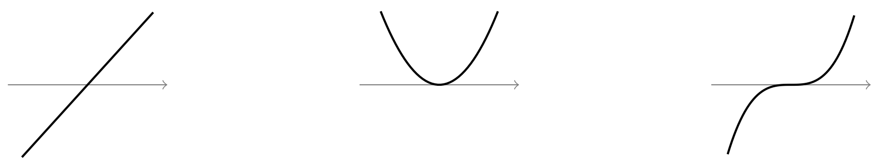
Zeros, Multiplicity, and Graph Behavior

Factoring to find zeros, reading multiplicity as crossing or touching, describing end behavior, and building a polynomial from its zeros

What multiplicity looks like on a graph. A factor's exponent tells you the shape of the curve where it meets the x-axis: an odd exponent passes through, an even exponent bounces off, and a higher odd exponent flattens as it passes.



odd exponent: crosses   even exponent: touches   higher odd: flattens

The three local shapes at a zero. No values shown: use the factors given in each problem.

Directions.

- Factor completely before reading anything off a polynomial — the factored form is what shows the zeros and their multiplicities.
- When a problem asks for behavior, name the zero, its multiplicity, and whether the curve crosses or touches.
- End behavior is set by the degree (odd or even) and the sign of the leading coefficient.

1. Factor $f(x) = x^3 - 4x$ completely, then list every real zero of f. Write the factored form on the answer line.

Answer: ________________

2. A polynomial is given in factored form:

$$g(x) = (x + 1)(x - 3)^2$$

List each real zero with its multiplicity, and state whether the graph of g crosses or touches the x-axis at each one. Write the zeros on the answer line.

Answer: ________________

3. Factor $h(x) = x^3 - 3x^2 - 10x$ completely, then list every real zero of h. Write the factored form on the answer line.

Answer: ________________

4. Describe the end behavior of $f(x) = -2x^3 + 5x - 1$ by completing both statements, and say in one sentence which two features of the polynomial decided them.

as $x \to \infty,\ f(x) \to$ _________ as $x \to -\infty,\ f(x) \to$ _________

Answer: __________________

5. A polynomial is given in factored form:

$$p(x) = x^2(x-4)^3$$

State the degree of p, list each real zero with its multiplicity, and say whether the graph crosses, touches, or flattens as it passes at each zero. Write the zeros on the answer line.

Answer: __________________

6. An engineer needs a curve that meets the x-axis exactly at $x = -2$ and at $x = 1$, passing straight through at $x = -2$ but bouncing off at $x = 1$. Write the polynomial of least degree with leading coefficient 1 that does this, in expanded form.

Answer: ________________

7. Solve $x^4 - 5x^2 + 4 = 0$ by factoring. (Hint: the left side is a quadratic in x^2.) List all four real zeros on the answer line.

Answer: ________________

8. Describe the end behavior of $g(x) = 3x^4 - x^3$ by completing both statements, and say in one sentence why both ends behave the same way here but not in problem 4.

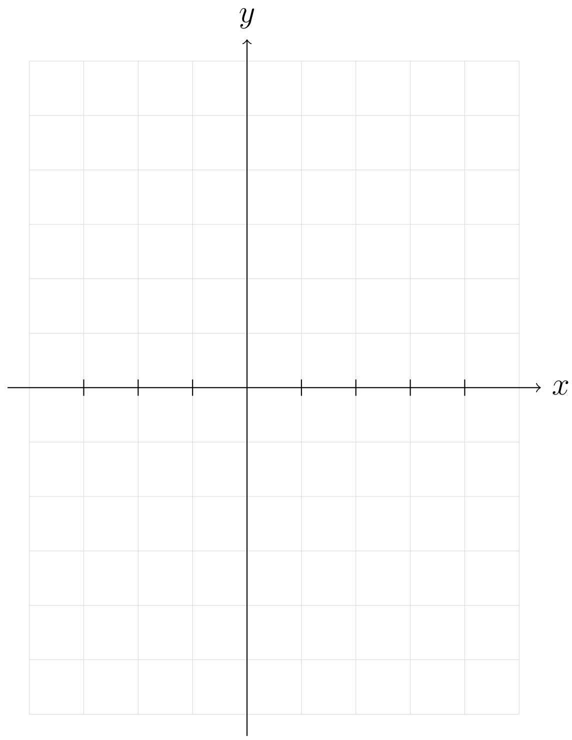
as $x \to \infty,\ g(x) \to$ ________ as $x \to -\infty,\ g(x) \to$ ________

Answer: ________________

9. A designer sketches a profile curve that crosses the x-axis at $x = -3$ and just touches it at $x = 2$, with no other x-intercepts. Write the polynomial of least degree with leading coefficient 1 that matches the sketch, in expanded form.

Answer: ________________

10. Sketch the graph of $y = x(x+2)^2(x-3)$ on the grid below. Mark every x-intercept, label each one as crossing or touching, and make both ends of your sketch match the end behavior. Each square is one unit wide; choose your own vertical scale and label it. Write the zeros on the answer line.



Answer: ________________